Characterization of Message Lengths for Capacity-Achieving Linear PIR

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***Abstract*—We characterize all positive message lengths of capacity-achieving linear private information retrieval (CALPIR) schemes with $M \geq 2$ replicated messages and $N \geq 2$ noncolluding servers. Over every finite field, the achievable lengths are precisely $a(N-1)+bN^{M-1}$, where a, b are nonnegative integers, not both zero. Necessity follows from the pointwise equality conditions of the PIR capacity converse and the integrality of linear conditional entropies. We also revisit T-private PIR for $2 \leq T < N$ and give a pointwise entropy proof that every capacity-achieving linear scheme satisfies $\gcd(N,T)(N/\gcd(N,T))^{M-1} \mid L$. Known constructions attain every positive multiple of this quantity over fields of cardinality at least N. The T-private divisibility threshold and large-field constructions are known from earlier subpacketization work; our treatment gives an explicit querywise entropy derivation under the stated download convention.**

I. Problem Setting and Related Work

Let $M, N \geq 2$ and $L \geq 1$ be integers, and let $\mathbb{F}_p$ be a finite field of cardinality p, where p is a prime power. We put $[M] := \{1, \ldots, M\}$ and $[N] := \{1, \ldots, N\}$. All entropies use logarithms to base p. The M messages $W^1, \ldots, W^M \in \mathbb{F}_p^L$ are independent and uniformly distributed, and are replicated at all N servers. The user selects an index θ, uniformly distributed on $[M]$, and draws queries $Q_n \in \mathcal{Q}_n$ for $n \in [N]$, where each query alphabet $\mathcal{Q}_n$ is finite. The answer of server n is A_n. Write $Q_{[N]} = (Q_1, \ldots, Q_N)$, $A_{[N]} = (A_1, \ldots, A_N)$, and $W^{[M]} = ((W^1)^\top, \ldots, (W^M)^\top)^\top$. The requirements are

1) $\mathrm{I}(\theta, Q_{[N]}; W^{[M]}) = 0$ (query independence);
2) $\mathrm{H}(A_n \mid Q_n, W^{[M]}) = 0$ for each n (deterministic answers);
3) $\mathrm{H}(W^m \mid A_{[N]}, Q_{[N]}, \theta = m) = 0$ for every $m \in [M]$ (zero-error correctness);
4) $\Pr(Q_n = q \mid \theta = m) = \Pr(Q_n = q \mid \theta = m')$ for every n, m, m', q (one-server privacy).

A scheme is *linear* if, for every n and $q \in \mathcal{Q}_n$, there is a matrix $\mathbf{G}_n^q$ over $\mathbb{F}_p$ such that $A_n = \mathbf{G}_n^{Q_n} W^{[M]}$. The number of rows of $\mathbf{G}_n^q$ may depend on q; thus the answer is a vector of query-dependent length. We measure the expected *entropy download* by

$$D := \sum_{n \in [N]} \mathrm{H}(A_n \mid Q_n). \tag{1}$$

For a fixed query, deleting linearly dependent answer coordinates preserves the information content and recoverability. Accordingly, in the linear setting D can also be implemented as the expected number of transmitted, linearly independent field symbols, with the query-dependent choice of a row-space basis known to the user. Without this reduction, the actual number of transmitted coordinates may exceed D.

The PIR capacity converse [1] states that

$$\frac{D}{L} \geq \sum_{j=0}^{M-1} N^{-j} = \frac{1 - N^{-M}}{1 - N^{-1}}. \tag{2}$$

We call a linear scheme attaining equality *CALPIR*. Sun and Jafar constructed capacity-achieving schemes with $L = N^{M-1}$ [3]; Tian, Sun, and Chen constructed such schemes with $L = N - 1$ and established the minimum linear message length [2]. Our question is not merely the minimum: we determine *every* positive L for which equality in (2) is possible.

For colluding servers, Zhang and Xu studied the smallest subpacketization of linear capacity-achieving T-private PIR and established a threshold $\gcd(N,T)(N/\gcd(N,T))^{M-1}$ [5]; divisibility arguments already appear in their proof. Xu and Zhang subsequently constructed schemes at the threshold over fields with cardinality at least N [6]. We explicitly allow query-dependent answer dimensions and use the conditional-entropy download convention. The pointwise proof below works directly in this formulation; we do not claim that this distinction alone establishes a new achievable-length result beyond the earlier subpacketization theory.

II. Equality Conditions for Capacity Achievement

We first state and prove the equality case of Han's inequality. This will be used to obtain conditional independence at each supported query.

Lemma 1 (Equality in Han's inequality)**.** *Let $X_1, \ldots, X_N, Z$ be discrete random variables with finite conditional entropies, and let $1 \leq T < N$. If*

$$\frac{1}{\binom{N}{T}} \sum_{|\mathcal{T}|=T} \mathrm{H}(X_{\mathcal{T}} \mid Z) = \frac{T}{N} \mathrm{H}(X_{[N]} \mid Z), \tag{3}$$

then $X_1, \ldots, X_N$ are mutually conditionally independent given Z.

Proof. For $0 \leq k \leq N$, put $F_0 = 0$ and

$$F_k = \frac{1}{\binom{N}{k}} \sum_{|\mathcal{T}|=k} \mathrm{H}(X_{\mathcal{T}} \mid Z) \quad (k \geq 1).$$

By the entropy chain rule, the increment $F_k - F_{k-1}$ is the average of $\mathrm{H}(X_i \mid X_{\mathcal{T}}, Z)$ over all $(\mathcal{T}, i)$ with $|\mathcal{T}| = k-1$ and $i \notin \mathcal{T}$. Coupling nested uniformly random subsets and using the fact that conditioning reduces entropy shows that these increments are nonincreasing in k. Equation (3) states that the average of the first T increments equals the average of all N increments. Since $T < N$, all increments must therefore be equal. Consequently,

$$\sum_{n\in[N]} \mathrm{H}(X_n \mid Z) = NF_1 = F_N = \mathrm{H}(X_{[N]} \mid Z).$$

Equality in entropy subadditivity is equivalent to mutual conditional independence given Z. □

The next lemma records the pointwise equality conditions needed below. Equalities in the averaged capacity converse pass to every supported query because the individual entropy gaps are nonnegative. For $T = 1$ these specialize to the familiar independence and interference properties in [2].

Lemma 2. *Suppose $1 \le T < N$ and the privacy condition holds for every set of at most T servers. If a PIR scheme attains the T-private capacity, then the following hold for any nonempty $J \subseteq [M]$, any $m \in [M]$, and any $\mathbf{q} = (q_1, \dots, q_N)$ for which $\Pr(Q_{[N]} = \mathbf{q} \mid \theta = m) > 0$:*

1) *If $m \in J$, the server answers are conditionally independent,*

$$\begin{aligned}&\mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^{[M]\setminus J}, \theta = m)\\ &\quad= \sum_{n\in[N]} \mathrm{H}(A_n \mid Q_n = q_n, W^{[M]\setminus J}, \theta = m). \end{aligned} \tag{4}$$

2) *If $m \notin J$, every T-server set spans the full answer information after the indicated messages are revealed: for every $\mathcal{T} \subseteq [N]$ with $|\mathcal{T}| = T$,*

$$\begin{aligned}&\mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^{[M]\setminus J}, \theta = m)\\ &\quad= \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}} = q_{\mathcal{T}}, W^{[M]\setminus J}, \theta = m). \end{aligned} \tag{5}$$

Proof. Fix an arbitrary ordering $m_1, \dots, m_M$ of $[M]$ and put $S_j := \{m_1, \dots, m_j\}$ for $1 \le j \le M$. We spell out the capacity converse, including the intermediate inequalities whose equality cases will be needed. By one-server privacy, query independence, deterministic answers, and entropy subadditivity,

$$\begin{aligned} D &= \sum_{n\in[N]} \mathrm{H}(A_n \mid Q_n, \theta = m_1)\\ &= \sum_{n\in[N]} \mathrm{H}(A_n \mid Q_{[N]}, \theta = m_1)\\ &\ge \mathrm{H}(A_{[N]} \mid Q_{[N]}, \theta = m_1)\\ &= L + \mathrm{H}(A_{[N]} \mid Q_{[N]}, W^{m_1}, \theta = m_1). \end{aligned} \tag{6}$$

The last equality follows from the chain rule, zero-error correctness, and the independence of messages and queries.

For every nonempty proper $S \subsetneq [M]$, $m \in S$, and $m' \notin S$, the standard T-private converse step is

$$\begin{aligned}&\mathrm{H}(A_{[N]} \mid Q_{[N]}, W^S, \theta = m)\\ &\quad\ge \frac{1}{\binom{N}{T}} \sum_{|\mathcal{T}|=T} \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}}, W^S, \theta = m)\\ &\quad= \frac{1}{\binom{N}{T}} \sum_{|\mathcal{T}|=T} \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}}, W^S, \theta = m')\\ &\quad\ge \frac{T}{N} \mathrm{H}(A_{[N]} \mid Q_{[N]}, W^S, \theta = m'). \end{aligned} \tag{7}$$

Indeed, an answer subset is determined by its local query and the messages. Since the messages are independent of the full query and θ, its conditional entropy is unchanged if $Q_{[N]}$ is replaced by $Q_{\mathcal{T}}$. The first inequality is entropy monotonicity under taking an answer subset. The middle equality uses T-privacy, and the final inequality is conditional Han's inequality.

Apply (7) with $S = S_j$, $m = m_j$, and $m' = m_{j+1}$ for $1 \le j < M$. Correctness gives

$$\begin{aligned}&\mathrm{H}(A_{[N]} \mid Q_{[N]}, W^{S_j}, \theta = m_j)\\ &\quad\ge \frac{T}{N} \mathrm{H}(A_{[N]} \mid Q_{[N]}, W^{S_j}, \theta = m_{j+1})\\ &\quad= \frac{T}{N} \big(L + \mathrm{H}(A_{[N]} \mid Q_{[N]}, W^{S_{j+1}}, \theta = m_{j+1})\big). \end{aligned} \tag{8}$$

The equality is exact: $W^{m_{j+1}}$ is independent of $(Q_{[N]}, W^{S_j}, \theta)$ and is recovered from the full answers, given the desired index and query. At the end of the recursion,

$$\mathrm{H}(A_{[N]} \mid Q_{[N]}, W^{S_M}, \theta = m_M) = 0$$

by answer determinism. Substituting (8) successively into (6) yields the full chain

$$D \ge L + \frac{T}{N} L + \cdots + \left(\frac{T}{N}\right)^{M-1} L. \tag{9}$$

If capacity is attained, the two sides of (9) coincide. Every inequality used above has a nonnegative gap and appears with a strictly positive coefficient. Hence equality holds in the initial subadditivity step and in *both* inequalities of (7), for every stage of this ordering. Since the ordering is arbitrary, the latter conclusion applies to every nonempty proper S, every $m \in S$, and every $m' \notin S$: choose an ordering whose first $|S|$ elements are S, with m last and m' immediately afterward.

We next pass from averaged equalities to individual supported queries. For a fixed $S, m, \mathcal{T}$, the difference

$$\begin{aligned}&\mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^S, \theta = m)\\ &\quad- \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}} = q_{\mathcal{T}}, W^S, \theta = m)\end{aligned}$$

is nonnegative for each $\mathbf{q}$ supported under $\theta = m$. The corresponding equality in the first line of (7) shows that its expectation is zero, for every T-subset $\mathcal{T}$. Thus the difference

vanishes for every supported $\mathbf{q}$. Similarly, for $m' \notin S$, the conditional Han gap

$$\frac{1}{\binom{N}{T}} \sum_{|\mathcal{T}|=T} \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}} = q_{\mathcal{T}}, W^S, \theta = m') - \frac{T}{N} \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^S, \theta = m')$$

is nonnegative and has zero expectation, so it vanishes at every $\mathbf{q}$ supported under $\theta = m'$. Lemma 1 then gives mutual conditional independence of the server answers at that query, conditioned on W^S.

Finally let J and m be as in the statement. If $J = [M]$, choose $m_1 = m$ in (6); its initial subadditivity equality holds at every supported query by the same nonnegative-gap argument. This gives (4). If $J \subsetneq [M]$ and $m \in J$, take $S = [M] \setminus J$. Choosing m immediately after S in the ordering and applying the pointwise Han equality with desired index m yields (4). If $m \notin J$, take $S = [M] \setminus J$, put m last among the elements of S, and choose the next index in J. The pointwise first-step equality gives (5) for every T-subset. □

III. Complete Classification for Noncolluding Servers

Define the positive part of the numerical semigroup

$$\mathrm{AM} := \{a(N-1) + bN^{M-1} : a, b \in \mathbb{Z}_{\geq 0}\} \setminus \{0\}. \tag{10}$$

Theorem 1. *For any finite field $\mathbb{F}_p$ and any integers $M, N \geq 2$, a CALPIR scheme with positive message length L exists if and only if $L \in \mathrm{AM}$.*

Proof. The constructions with message lengths $N-1$ and N^{M-1} exist over every finite field [2], [3]. Direct sums preserve linearity, correctness, privacy, and equality in the normalized download bound, proving sufficiency.

For necessity fix $m \in [M]$ and a supported query vector $\mathbf{q}$. Write

$$h(\mathbf{q}) := \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^m, \theta = m).$$

Take $T = 1$. By Lemma 2 with $J = [M]$ and $J = [M] \setminus \{m\}$, respectively, and by correctness,

$$\begin{aligned} N h(\mathbf{q}) &= \sum_{n \in [N]} \mathrm{H}(A_n \mid Q_{[N]} = \mathbf{q}, W^m, \theta = m) \\ &\leq \sum_{n \in [N]} \mathrm{H}(A_n \mid Q_{[N]} = \mathbf{q}, \theta = m) \\ &= \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, \theta = m) \\ &= L + h(\mathbf{q}). \end{aligned} \tag{11}$$

The last identity is the chain rule using the zero-error recovery of W^m and query independence. Since the scheme is linear, $h(\mathbf{q})$ is the rank of the answer map restricted to the $M-1$ undesired message blocks and hence is an integer. Thus

$$h(\mathbf{q}) \leq \left\lfloor \frac{L}{N-1} \right\rfloor. \tag{12}$$

Averaging over supported queries and using the equality $D = \mathrm{H}(A_{[N]} \mid Q_{[N]}, \theta = m)$ from the converse,

$$\begin{aligned} D - L &= \mathrm{H}(A_{[N]} \mid Q_{[N]}, W^m, \theta = m) \\ &\leq \left\lfloor \frac{L}{N-1} \right\rfloor. \end{aligned} \tag{13}$$

Capacity attainment and (2) give

$$L - \frac{L}{N^{M-1}} \leq (N-1) \left\lfloor \frac{L}{N-1} \right\rfloor. \tag{14}$$

Let b be the least nonnegative residue of L modulo $N-1$. Then (14) is equivalent to $bN^{M-1} \leq L$. Since $N^{M-1} \equiv 1 \pmod{N-1}$,

$$a := \frac{L - bN^{M-1}}{N-1} \in \mathbb{Z}_{\geq 0}.$$

Consequently $L = a(N-1) + bN^{M-1} \in \mathrm{AM}$. □

A. Consequences of the Classification

For $N = 2$, $\mathrm{AM} = \mathbb{Z}_{\geq 1}$. If $N \geq 3$, the two semigroup generators are coprime. The Frobenius coin theorem then shows that the largest positive length outside AM is

$$(N-2)(N^{M-1} - 1) - 1, \tag{15}$$

except when $(M, N) = (2, 3)$, for which (15) still equals 1 but the interval considered below is empty. There are exactly

$$\frac{(N-2)(N^{M-1}-1)}{2} \tag{16}$$

nonachievable positive lengths, and every $L \geq (N-2)(N^{M-1}-1)$ is achievable. For $N \geq 3$ and $(M, N) \neq (2, 3)$, the number of achievable lengths in $[N-1, (N-2)(N^{M-1}-1) - 1]$ is

$$\frac{(N-2)(N^{M-1}-1)}{2} - 1. \tag{17}$$

For $(M, N) = (2, 3)$, $\mathrm{AM} = \mathbb{Z}_{\geq 2}$.

IV. Extension to T-Private PIR

Suppose $2 \leq T < N$. Privacy is now required for every $\mathcal{T} \subseteq [N]$ with $|\mathcal{T}| \leq T$:

$$\begin{aligned} &\Pr(Q_{\mathcal{T}} = q_{\mathcal{T}} \mid \theta = m) \\ &\quad = \Pr(Q_{\mathcal{T}} = q_{\mathcal{T}} \mid \theta = m') \end{aligned} \tag{18}$$

for all $m, m' \in [M]$ and all $q_{\mathcal{T}}$. The capacity bound is [4]

$$\frac{D}{L} \geq \sum_{j=0}^{M-1} (T/N)^j. \tag{19}$$

Theorem 2. *Every capacity-achieving linear T-private PIR scheme over any finite field $\mathbb{F}_p$ has a message length satisfying*

$$\gcd(N, T) \left(\frac{N}{\gcd(N, T)} \right)^{M-1} \mid L. \tag{20}$$

Conversely, for every $p \geq N$ and every positive L satisfying (20)*, such a scheme exists over $\mathbb{F}_p$.*

The numerical threshold in (20) and the large-field construction are known from [5], [6]. The argument below proves the divisibility condition directly with pointwise conditional entropy equalities, allowing the answer dimension to depend on the query. No distinct novelty of the numerical threshold is asserted.

Lemma 3. *Fix a nonempty proper $J \subsetneq [M]$. For every $m \in [M]$ and every $\mathbf{q}$ supported given $\theta = m$, the values*

$$\mathrm{H}(A_n \mid Q_n = q_n, W^{[M]\setminus J}, \theta = m), \quad n \in [N], \tag{21}$$

are equal.

Proof. Put $r_n(q_n) := \mathrm{H}(A_n \mid Q_n = q_n, W^{[M]\setminus J})$. Query independence implies that this quantity is independent of the requested index when q_n is held fixed. Choose $m' \in J$ and $m'' \notin J$. For any supported $\mathbf{q}'$ given $\theta = m''$ and any T-subset $\mathcal{T}$, privacy supplies a query vector supported for $\theta = m'$ with the same restriction $q'_{\mathcal{T}}$. By the two parts of Lemma 2,

$$\sum_{n\in\mathcal{T}} r_n(q'_n) = \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}', W^{[M]\setminus J}, \theta = m''). \tag{22}$$

The right-hand side is independent of $\mathcal{T}$. Comparing two T-subsets differing in one server index shows that all $r_n(q'_n)$ are equal because $T < N$. Finally, given an arbitrary supported $\mathbf{q}$ and distinct n, n', a T-subset containing n, n' exists since $T \geq 2$. Privacy supplies $\mathbf{q}'$ supported for $\theta = m''$ with the same local query values at those two servers. Hence $r_n(q_n) = r_{n'}(q_{n'})$. □

Proof of Theorem 2. Let $1 \leq s \leq M-1$, choose $J \subseteq [M]$ with $|J| = s$, and $m \in J$. For any supported full query $\mathbf{q}$ we prove, by induction on s, that each server has conditional entropy

$$\mathrm{H}(A_n \mid Q_n = q_n, W^{[M]\setminus J}, \theta = m) = \frac{L}{N^s} \sum_{j=0}^{s-1} N^{s-1-j} T^j. \tag{23}$$

For $s = 1$, conditioning on every message except the desired one and using correctness yields total conditional entropy L at each supported full query. Equation (4) and Lemma 3 then give L/N per server.

For $s \geq 2$, select $m' \in J \setminus \{m\}$, a T-subset $\mathcal{T}$, and a supported query $\mathbf{q}'$ given $\theta = m'$ with $q'_{\mathcal{T}} = q_{\mathcal{T}}$, as guaranteed by privacy. Using correctness, the two parts of Lemma 2, and the induction hypothesis, we obtain

$$\begin{aligned} &N\mathrm{H}(A_n \mid Q_n = q_n, W^{[M]\setminus J}, \theta = m) \\ &= \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^{[M]\setminus J}, \theta = m) \\ &= L + \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^{[M]\setminus (J\setminus\{m\})}, \theta = m) \\ &= L + \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}} = q_{\mathcal{T}}, W^{[M]\setminus (J\setminus\{m\})}, \theta = m) \\ &= L + \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}} = q'_{\mathcal{T}}, W^{[M]\setminus (J\setminus\{m\})}, \theta = m') \\ &= L + \frac{TL}{N^{s-1}} \sum_{j=0}^{s-2} N^{s-2-j} T^j. \end{aligned} \tag{24}$$

This proves (23). The left-hand side of (23) is the rank of a fixed linear map over $\mathbb{F}_p$ and therefore an integer. In particular, $s = 1$ gives $N \mid L$. Taking $s = M-1$ gives

$$\frac{L}{N^{M-1}} \sum_{j=0}^{M-2} N^{M-2-j} T^j \in \mathbb{Z}. \tag{25}$$

Since $N \mid L$ and

$$\gcd\left(\left(\frac{N}{\gcd(N,T)}\right)^{M-2}, \sum_{j=0}^{M-2} \left(\frac{N}{\gcd(N,T)}\right)^{M-2-j} \left(\frac{T}{\gcd(N,T)}\right)^{j}\right) = 1, \tag{26}$$

(25) implies $N(N/\gcd(N,T))^{M-2} \mid L$, which is (20).

For sufficiency, Xu and Zhang constructed a linear capacity-achieving T-private scheme with length equal to the left-hand divisor of (20) over every finite field with $p \geq N$ [6]. Direct sums yield every positive multiple, without changing the field. □

For $p < N$, scalar extension yields a more limited sufficient condition. If $p^s \geq N$, a scheme over $\mathbb{F}_{p^s}$ can be viewed as a scheme over $\mathbb{F}_p$ after expressing each field symbol in an $\mathbb{F}_p$-basis. This proves achievability for positive lengths divisible by

$$s \gcd(N,T) \left(\frac{N}{\gcd(N,T)}\right)^{M-1}.$$

The sufficiency of (20) itself over all smaller fields is not asserted here.

V. Conclusion

For noncolluding servers, the set of all positive message lengths of capacity-achieving linear PIR is the numerical semigroup generated by $N-1$ and N^{M-1}, with no restriction on the finite-field size. For $2 \leq T < N$, a pointwise entropy argument recovers the known subpacketization divisibility condition under our query-dependent answer-dimension and entropy-download convention; matching constructions are available when the field cardinality is at least N. The two cases underscore how noncolluding PIR can admit asymmetric short-message constructions that do not persist under stronger collusion privacy.

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